

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	P. Lohith Krishna	Reg. No.:	192325122
Date:	19/08/2026	Duration:	60 mins

Code of Conduct

I P. Lohith Krishna / 192325122 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Lohith Krishna

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, design and document comprehensive test cases to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the functional and non-functional requirements to be tested.
2. Identify the test scenarios for the assigned problem statement.
3. Prepare detailed test cases covering normal, boundary, invalid, and exceptional conditions.
4. Include Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status.
5. Apply appropriate test design techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate requirement and functional coverage for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Smart Campus Maintenance and Service Request Management System

1. Introduction

The Smart Campus Maintenance and Service Request Management System is a web and mobile application designed to help students, faculty, and staff report campus service issues and track their resolution efficiently. This Test Case Design Document outlines the testing strategies, scenarios, and detailed test cases required to ensure the system is functionally robust, secure, reliable, and user-friendly.

2. Scope

This document covers functional and non-functional testing of the Smart Campus Maintenance and Service Request Management System. The primary modules in scope include User Authentication, Service Request Creation, Issue Categorization, Request Assignment, Status Tracking, Notifications, and Service Reports. Out of scope are physical maintenance work, hardware repair activities, and third-party messaging provider internal testing.

3. Functional Requirements

- FR01: Users shall be able to register and log in using an institutional email address and password.
- FR02: Users shall be able to create a maintenance request by selecting a category, location, priority, and description.
- FR03: Users shall be able to attach supporting images or documents to a service request.
- FR04: Administrators shall be able to assign service requests to maintenance staff and update their status.
- FR05: Users shall be able to view request status, receive updates, and close a request after resolution.

4. Non-Functional Requirements

- NFR01: The system shall support at least 3,000 concurrent users without significant performance degradation.
- NFR02: Service request submission and dashboard loading shall complete within 3 seconds under normal load.
- NFR03: User passwords and submitted personal information shall be stored using secure encryption/hashing practices.
- NFR04: The application interface shall be responsive and usable on common mobile, tablet, and desktop browsers.

5. Test Scenarios

- TS_01: Verify User Registration and Authentication functionality.
- TS_02: Verify Service Request Creation, Validation, and Attachment Upload.
- TS_03: Verify Request Assignment, Priority Handling, and Status Tracking.
- TS_04: Verify Notifications, Resolution Confirmation, and Request Closure.

6. Test Design Techniques

To ensure optimal test coverage, several black-box test design techniques are utilized:

- Equivalence Partitioning (EP): Used for validating request priority, issue categories, and mandatory field combinations.
- Boundary Value Analysis (BVA): Applied to attachment size limits and description length limits.
- Decision Table Testing: Used for combinations of user role, request priority, and assignment/status conditions.
- State Transition Testing: Used to model the lifecycle of a request (Submitted -> Assigned -> In Progress -> Resolved -> Closed).

7. Test Data

- Valid User Credentials: student@simats.edu.in / Campus@123
- Invalid User Credentials: student@simats.edu.in / Wrong@123
- Valid Request: Category 'Electrical', Location 'Block A - Lab 2', Priority 'High', Description 'Ceiling light not working'.
- Attachment Data: JPG image, 2 MB; boundary file: 10 MB.

7. Test Data (Continued)

- Invalid Request Data: Blank category, missing location, and empty description.
- Boundary Description Data: 1 character and 500 characters.
- Assignment Data: Staff ID MS014, Request ID SCR1025.
- Notification Data: Valid registered email and in-app notification enabled.

8. Detailed Test Cases

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_SCMS_001	User Registration (Valid)	1. Open Registration. 2. Enter valid institutional email and password. 3. Submit.	student@simats.edu.in Campus@123	Account is created and confirmation is shown.	Account created successfully.	Pass
TC_SCMS_002	User Login (Invalid)	1. Open Login. 2. Enter incorrect password. 3. Click Login.	student@simats.edu.in Wrong@123	System displays 'Invalid credentials'.	Invalid credentials displayed.	Pass
TC_SCMS_003	Create Service Request	1. Open New Request. 2. Select category/location/priority. 3. Enter description. 4. Submit.	Electrical; Block A-Lab 2; High	Request is created with unique ID and Submitted status.	Request SCR1025 created.	Pass
TC_SCMS_004	Mandatory Field Validation	1. Open New Request. 2. Leave location blank. 3. Submit.	Category: Plumbing Location: blank	System prevents submission and highlights required field.	Location field highlighted.	Pass

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_SCMS_005	Attachment Size Boundary	1. Add valid request. 2. Upload 10 MB image. 3. Submit.	JPG image, 10 MB	System accepts file at maximum permitted size.	10 MB file accepted.	Pass

8. Detailed Test Cases (Continued)

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_SCMS_006	Assignment to Staff	1. Login as admin. 2. Open pending request. 3. Assign staff. 4. Save.	Request SCR1025 Staff MS014	Request is assigned and staff receives notification.	Request assigned; notification sent.	Pass
TC_SCMS_007	Status Transition	1. Open assigned request. 2. Change status to In Progress. 3. Change to Resolved.	SCR1025	Status changes follow allowed sequence.	Status updated correctly.	Pass
TC_SCMS_008	Unauthorized Update	1. Login as student. 2. Open another user's request. 3. Attempt to change priority.	Student account SCR2044	System denies modification and keeps original data.	Update denied.	Pass
TC_SCMS_009	Resolution Closure	1. Open resolved request. 2. Confirm resolution. 3. Close request.	SCR1025	Request becomes Closed and closure time is recorded.	Request closed successfully.	Pass
TC_SCMS_010	Invalid Attachment	1. Create request. 2. Upload unsupported executable file. 3. Submit.	setup.exe	System rejects unsafe file and shows validation message.	Unsupported file rejected.	Pass
TC_SCMS_011	Performance – Dashboard	1. Simulate 3,000 users. 2. Open service dashboard. 3. Measure response time.	3,000 concurrent users	Dashboard remains available and loads within 3 seconds.	Dashboard loaded within target.	Pass

9. Decision Table – Request Assignment and Status

Condition / Rule	Rule 1	Rule 2	Rule 3	Rule 4
User Logged In?	Yes	Yes	No	Yes
Admin/Staff Role?	Yes	No	N/A	Yes
Request Assigned?	Yes	Yes	N/A	No
Action: Update Status	Yes	Limited	No	No
Action: Display Error	No	No	Yes (Login Req)	Yes (Assignment Req)

10. State Transition Testing – Service Request

Current State	Event / Action	Next State
Submitted	Administrator assigns staff	Assigned
Assigned	Staff starts work	In Progress
In Progress	Staff marks issue resolved	Resolved
Resolved	User confirms resolution	Closed
In Progress	Request cancelled by authorized user	Cancelled

11. Requirement Traceability and Coverage

Requirement ID	Requirement Description	Mapped Test Case IDs
FR01	User Registration/Login	TC_SCMS_001, TC_SCMS_002
FR02	Service Request Creation	TC_SCMS_003, TC_SCMS_004, TC_SCMS_014
FR03	Attachment Upload	TC_SCMS_005, TC_SCMS_010
FR04	Assignment and Status Tracking	TC_SCMS_006, TC_SCMS_007, TC_SCMS_008
FR05	Resolution, Notification and Closure	TC_SCMS_009, TC_SCMS_012, TC_SCMS_013

12. Coverage Summary

The test cases designed above provide full functional coverage of the five core functional requirements identified for the system. Boundary cases, invalid inputs, authorization checks, performance expectations, notifications, and logical state transitions have been included to improve overall quality and reduce production risk.

13. Entry and Exit Criteria

Entry Criteria:

- The test environment must be configured and accessible.
- All test data and user accounts must be prepared and loaded.
- The application build must be deployed successfully without critical build errors.

Exit Criteria:

- 100% of planned test cases must be executed.
- All Critical and High severity defects must be resolved and verified.
- No Priority 1 defects should remain open.

14. Conclusion

This Test Case Design Document provides a structured and comprehensive testing approach for the Smart Campus Maintenance and Service Request Management System. Through the application of Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing, the major functional and quality risks are addressed. The defined test cases provide strong coverage and support reliable deployment of the system for end users.

Rubrics for Calculation				
Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification (15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness (25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited/incomplete
Test Data & Expected Results (20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques (15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Criteria				
Criteria			Mark	
Requirement & Test Scenario Identification (15%)			/5	
Test Case Design & Completeness (25%)			/4	
Test Data & Expected Results (20%)			/4	
Application of Test Design Techniques (15%)			/4	
Testability & Test Coverage (15%)			/4	
Report Writing & Presentation (10%)			/4	
TOTAL			/25	